

Lecture 1 · Preference, Choice and Demand

MWG Chapters 1 and 2 · two goals, one hypothesis — and what each assumption buys

Advanced Microeconomics 1

Monday 21 September

Before any economics: twenty minutes of decisions

The Session 1 quiz

- Open the course page and follow the **quiz link** [link / QR here]. Work alone, in silence; about 20 minutes.
 - Around thirty short questions: allocation problems and a few quick choices. **There are no right answers** — choose what you actually prefer.
 - You will invent a **random ID** on the first screen — *write it down*, you will need it again. Responses are anonymous, ungraded, and you may skip the whole thing without penalty. [Optional: one decision, for one randomly chosen participant, will be paid for real.]
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- Your answers become a dataset that this course will return to more than once this term. That is all I will say about it today.

Two things economists are for

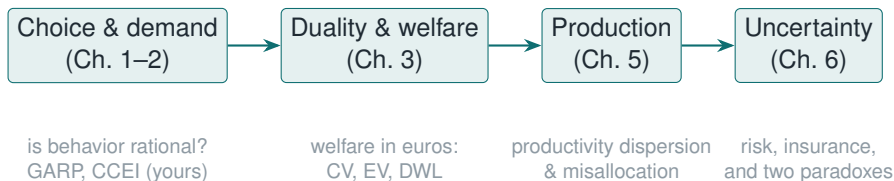
The frame for the whole course

Academic economists do two things: **(a)** try to understand how the world works, and **(b)** design policy that makes people better off. The formal tools have one job: to tell you exactly what you can learn from data, under exactly which assumptions.

For consumer behavior, the goals sharpen into two questions:

- (a) Is observed choice behavior rational?** (*A testable claim, not a creed: you test it on yourselves a week Thursday.*)
- (b) When prices or incomes change — a tax, a subsidy, inflation — who is better off, who is worse off, and by how much?**
 - The route from (a) to (b) runs through a theorem: welfare in *your* interest requires recovering *your* preferences from *your* behavior — possible only if (a)'s answer is yes.
 - **Assumptions are the currency.** Each one you grant buys sharper conclusions

This course in one arc



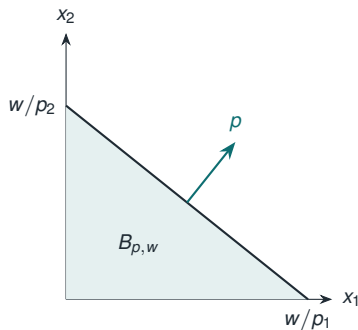
- Each block runs the same loop: state the two goals, buy assumptions, test (a), execute (b), then price the assumptions with evidence — some of it generated in this room twenty minutes ago.
- Problem sets stay ruthlessly technical: the applications are why you tolerate the algebra, not a substitute for it.

Formalise: demands, prices, wealth

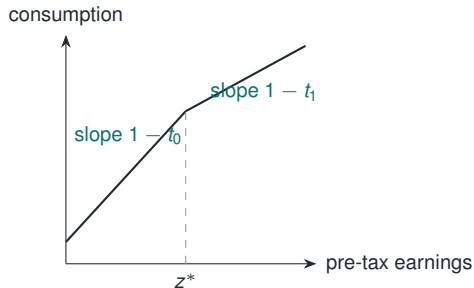
- L commodities, distinguished by traits, *location, date, and state of the world* — one definition that carries us quietly to GE and finance. Consumption set $X = \mathbb{R}_+^L$ (convex; rules out indivisibilities and survival constraints).
- **Walrasian budget set**, $p \gg 0$, $w > 0$:

$$B_{p,w} = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}.$$

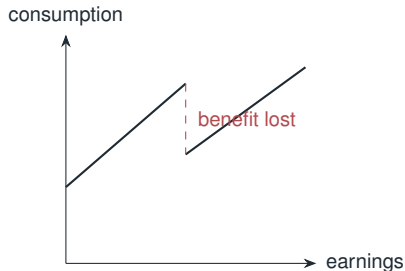
- Price taking; linearity (every unit at the same price); boundary a hyperplane with normal p ; convex — inherited by everything downstream.



Kink (e.g. progressive tax)



Notch (benefit cliff)



- Progressive taxes, EITC phase-outs, benefit cliffs: piecewise-linear, sometimes non-convex budget frontiers.
- Hold the thought: choice theory predicts *exactly what happens at these corners*, and that prediction becomes an identification strategy for elasticities in Lecture 4.

Definition · Walrasian demand

$x(p, w) \subseteq B_{p,w}$: the chosen bundle(s) at each price–wealth pair; assume single-valued and continuous for now.

Two assumptions maintained throughout Ch. 2 (both *derived* from preferences in Ch. 3):

1. **Homogeneity of degree zero:** $x(\alpha p, \alpha w) = x(p, w)$ — only relative prices and real wealth matter; “no money illusion.” Practical use: normalize.
2. **Walras’ law:** $p \cdot x(p, w) = w$ — full expenditure.
 - Differentiating the two identities yields free cross-equation restrictions on any demand system — **Engel aggregation** ($\sum_{\ell} p_{\ell} \partial x_{\ell} / \partial w = 1$: a marginal dollar is spent somewhere, so not every good can be inferior) and **Cournot aggregation**. You derive both on PS1 and verify them on CES.

The descriptive layer: commodities

What the formalisation already buys: describing commodities the descriptive layer

Before any rationality assumption, $x(p, w)$ supports a working vocabulary — most of applied demand description:

- **Wealth responses.** Engel curves $w \mapsto x_\ell(p, w)$; *normal* vs. *inferior*, *luxury* vs. *necessity* (wealth elasticity ≥ 1). Engel's law — the food share falls with wealth — is the oldest empirical regularity in economics, and still holds in every household survey.
- **Price responses.** Own-price: ordinary vs. *Giffen* — a logical possibility the theory refuses to rule out, and found in the field among subsidized rice consumers in Hunan (Jensen–Miller, AER 2008: a *policy experiment*).
- The two aggregation identities from the last slide (Engel, Cournot) discipline any estimated demand system for free — EC1 makes you derive and verify them.

An IOU, priced on Monday

L1: Preference, Choice & Demand / Advanced Micro / (JMWG) 10/24
“Substitutes” from raw cross-price signs is **incoherent**: $\partial x_1 / \partial p_2$ and $\partial x_2 / \partial p_1$ can

Goal (a): the rationality hypothesis

Why start with axioms? Because rationality is measurable

goal (a): understand

- “Rational agent” sounds like an article of faith. It is not: consistency with utility maximization is a **testable, quantifiable property** of a finite dataset of choices.
- Humans: in a large representative Dutch panel, consistency scores are heterogeneous and correlate systematically with wealth, education, and age (Choi, Kariv, Müller, Silverman, AER 2014).
- (In passing: run on GPT, the same test scores the machine *more* consistent than most humans — Chen et al., PNAS 2023.)
- CKMS is **this block’s paper for goal (a)**: by Thursday week you will have replicated its design on yourselves. The machinery starts now.

Primitives and two routes

Let X be the set of alternatives (bundles, careers, portfolios, ...).

Route A: preference-based

Start from a preference relation $\succsim$ on X ; impose *rationality axioms*; derive choice behavior.

Route B: choice-based

Start from observed choice behavior itself; impose *consistency axioms* (WARP); ask when it looks “as if” preferences exist.

- Route B treats the model's testable implications as primitives — the ancestor of modern revealed-preference econometrics.
- Central question of Ch. 1: **when are the two routes equivalent?**

Definition · preference relation

$\succsim$ is a binary relation on X ; $x \succsim y$ reads “ x is at least as good as y .” Derived:
 $x \succ y \iff x \succsim y$ and not $y \succsim x$; $x \sim y \iff x \succsim y$ and $y \succsim x$.

Definition · rationality (MWG 1.B.1)

$\succsim$ is *rational* if it is **complete** ($x \succsim y$ or $y \succsim x$, always) and **transitive** ($x \succsim y \succsim z \Rightarrow x \succsim z$).

- What they rule out: incomparability (completeness — strong for unfamiliar or morally loaded choices); cycles and framing effects (transitivity). An agent with cyclic strict preferences can be **money-pumped**: pay ε at each step of the cycle, forever.
- Consequences (Prop. 1.B.1): $\succ$ irreflexive and transitive; $\sim$ an equivalence relation. Proofs: PS1.

Definition · utility representation

$u : X \rightarrow \mathbb{R}$ represents $\succsim$ if $x \succsim y \iff u(x) \geq u(y)$.

- If some u represents $\succsim$, then $\succsim$ is rational (completeness and transitivity of $\geq$ transfer back).
- **Ordinality:** any strictly increasing $f \circ u$ represents $\succsim$ too. Levels carry no meaning; only the ranking does. Anything not invariant to monotone transformations is not (yet) economics — cardinal content returns only under uncertainty (Ch. 6).
- **Existence:** finite X — rationality suffices ($u(x) = \#\{y : x \succsim y\}$, PS1). Infinite X — rationality is *not* enough: lexicographic preferences on $\mathbb{R}^2$ are rational, strongly monotone, and have **no representation**. The missing ingredient is continuity; Debreu delivers existence next lecture.

Definition · choice structure

$(\mathcal{B}, C(\cdot))$: a family $\mathcal{B}$ of budget sets $B \subseteq X$ and nonempty chosen sets $C(B) \subseteq B$. Read: **the experiments we ran, and their outcomes**. No mental states assumed — which is why this travels to lab data, scanner data, and LLM outputs.

Definition · WARP (MWG 1.C.1)

If $x, y \in B$ with $x \in C(B)$, then for any B' with $x, y \in B'$ and $y \in C(B')$, also $x \in C(B')$.

- In words: if x is ever chosen with y available, y may never be chosen *over* x .
- Violation: $C(\{x, y\}) = \{x\}$ but $C(\{x, y, z\}) = \{y\}$. Not exotic — exactly what decoys, menu effects, and limited attention produce. WARP is where standard and behavioral economics part ways.

Proposition · MWG 1.D.1

If $\succsim$ is rational, the induced choice rule $C^*(B, \succsim) = \{x \in B : x \succsim y \ \forall y \in B\}$ satisfies WARP.

Proof. $x \in C^*(B)$, $y \in B$, and $y \in C^*(B')$, $x \in B'$. Then $x \succsim y$ and $y \succsim z$ for all $z \in B'$; transitivity chains to $x \succsim z$ for all $z \in B'$, so $x \in C^*(B')$. $\square$

Proposition · MWG 1.D.2

If $(\mathcal{B}, C)$ satisfies WARP **and** $\mathcal{B}$ contains all subsets of up to three elements, then the revealed relation $\succsim^*$ ($x \succsim^* y$ iff x chosen with y available) is the unique rational preference rationalizing C .

- Why *three*? Transitivity is a statement about triples; with pairs only, WARP can hold while no rational preference exists (PS1 makes you build the example).

The empirical problem

Real consumption data give us only *budget sets* — a very poor $\mathcal{B}$. Pairs and triples are never observed, so WARP alone will not secure rationalizability in the market setting.

- Fix: strengthen WARP to rule out revealed-preference *chains* cycling back — **GARP**: no sequence $x^1 R x^2 R \dots R x^n$ with x^n strictly revealed preferred to x^1 .
- **Afriat's Theorem** (1967; stated and proved in Lecture 4): a finite dataset $\{(p^t, x^t)\}$ satisfies GARP *iff* it is rationalizable by a continuous, increasing, concave utility. For finite data, **utility maximization = GARP** — nothing more, nothing less.
- Thursday week's practical: you implement this equivalence, on real choices, before you ever see it proved.

Goal (a): the test for market data

Definition · WARP in the market (MWG 2.F.1)

$$p \cdot x(p', w') \leq w \text{ and } x(p', w') \neq x(p, w) \implies p' \cdot x(p, w) > w'.$$

- Read: if the bundle chosen at (p', w') was *affordable* at (p, w) but not chosen there, then when it *was* chosen, the other bundle must have been unaffordable. Otherwise the two observations contradict each other.
- One axiom about pairs of observations. Its power is unlocked by asking a well-chosen pair — next slide.

Definition · compensated price change

From (p, w) with $x = x(p, w)$, move to p' and set $w' = p' \cdot x$: wealth adjusted so *the old bundle remains exactly affordable* — the budget line pivots through the original choice, stripping out the mechanical income effect *using only observables*.

Proposition · MWG 2.F.1

Under h.d.0 and Walras' law: $\text{WARP} \iff$ for every compensated change,

$$(p' - p) \cdot [x(p', w') - x(p, w)] \leq 0, \quad \text{strictly if } x(p', w') \neq x(p, w).$$

Proof ($\Rightarrow$), the four lines to know cold. With $x' \neq x$:

$$(p' - p) \cdot (x' - x) = \underbrace{p' x'}_{=w'} - \underbrace{p' x}_{=w'} - p x' + \underbrace{p x}_{=w} = w - p x'.$$

Take dp small, compensate $dw = x \cdot dp$; the law of demand becomes $dp \cdot dx \leq 0$ with $dx = [D_p x + D_w x x^T] dp$.

Definition · Slutsky substitution matrix

$$S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^T, \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k.$$

- Every entry is estimable from demand data alone — **no utility function anywhere**.
- Why build it? Goal (a) needs a test for *both* data regimes: 25 observed choices $\rightarrow$ GARP (the practical); an estimated demand *function* $\rightarrow$ restrictions on S . Same question, two kinds of data. Built here from consistency; rebuilt from preferences on Monday — and the comparison is the deep structure of the block.

Proposition · consequences (MWG 2.F.2–2.F.3)

If x is h.d.0, satisfies Walras' law and WARP, then $S(p, w)$ is

- **negative semidefinite**: $dp \cdot S dp \leq 0$;
- **singular**: $S p = 0$ and $p^T S = 0$ (from homogeneity + Walras).

What WARP does not buy

For $L \geq 3$, WARP does **not** imply S **symmetric** (MWG Example 2.F.1). Symmetry is the extra content of *transitivity* — of one coherent preference behind all choices — delivered by full preference maximization in Lecture 3, and the key to recovering preferences from demand in Lecture 4.

- Keep the ledger: NSD + singularity = pairwise consistency. Symmetry = no cycles anywhere.

Scorecard, takeaways, readings

Takeaways

- **Goal (a):** hypothesis stated (rational $\succsim$ / utility); one test live for finite rich domains (WARP + triples), the market-data version flagged (GARP, Thursday week); for demand functions, S is NSD + singular from consistency alone — *symmetry conspicuously missing*. That gap is Monday's headline.
- **Goal (b):** not yet licensed — welfare needs recovered preferences, and recovery needs (a) to hold. The kinked budget sets are (b)'s down payment.

Required: MWG Ch. 1 and Ch. 2 (all); Example 2.F.1 carefully.

This block's papers: (a) Choi, Kariv, Müller & Silverman (AER 2014) — intro + Table 1 now, the rest before the practical. (b) Hausman (AER 1981) — skim now, read before Lecture 4. *The third paper of the block is revealed in Lecture 4.*

In passing today: Chen et al. (2023); Jensen–Miller (2008); Saez (2010).

Ahead: the practical is a week Thursday — all you need is a laptop with a working R installation.

Check it starts this week, not that morning